Ai Based Sign Language To Audio Converter

T05

Our Team

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Introduction

- SignBridge is an AI-based desktop application designed to translate sign language gestures into real-time text and spoken output.
- The system facilitates effective communication between sign language users and hearing individuals in digital environments.
- It supports accessibility and inclusivity in online meetings, virtual classrooms, and professional interactions.

Problem Statement

Communication Barrier:
Sign language users face challenges while communicating with hearing individuals in digital environments.
This limits effective participation in online interactions.

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Limitations of Existing Solutions:
Human interpreters and manual methods are not always available or scalable.
They increase dependency and reduce real-time accessibility.

Need for an Automated System:
An AI-based solution is required for real-time sign language translation.
This ensures inclusive and independent communication for users.

Goals Of Project

- Enhance Accessibility: Enable effective communication for sign language users through real-time text and voice translation.
- Ensure Real-Time Performance: Deliver fast and accurate gesture recognition with minimal latency.
- Promote Inclusivity: Reduce dependency on human interpreters in digital communication.
- Maintain Privacy: Perform all processing locally without storing or sharing user data.

Tech Stack

Frontend

- Desktop GUI using PyQt / Tkinter for live gesture feedback and controls

Backend

- Python execution engine
- OpenCV for real-time video capture and preprocessing
- MediaPipe Hands for 3D hand landmark detection

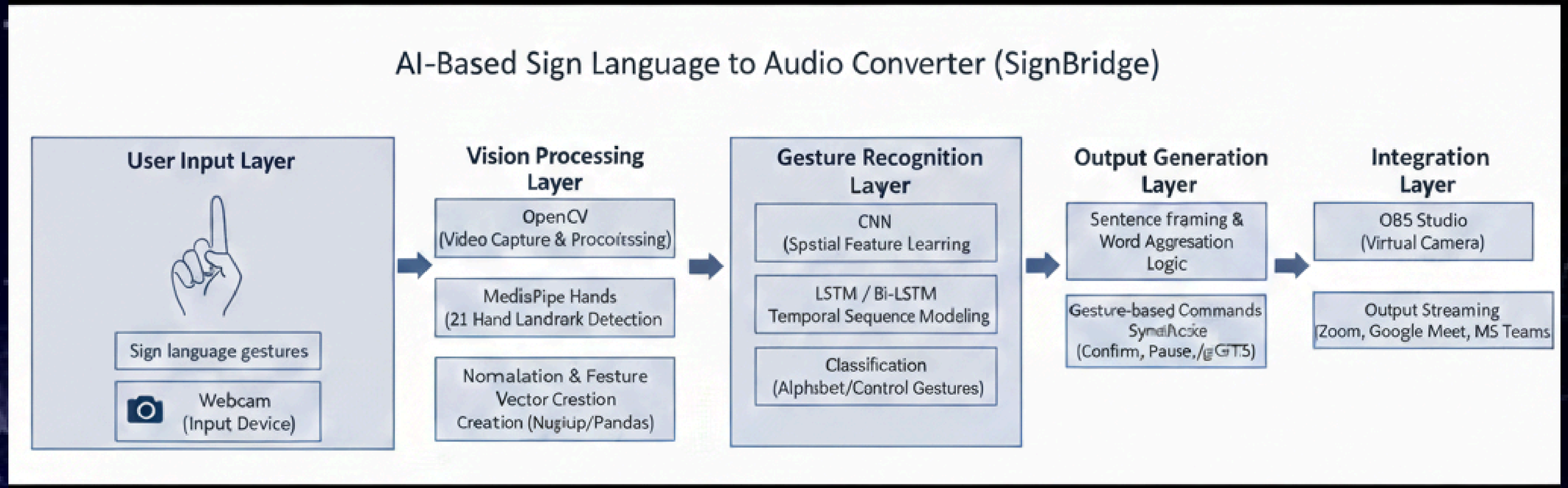
Model Layer (Advanced)

- CNN-based spatial feature extraction
- LSTM / Bi-LSTM for temporal gesture sequence modeling
- Hybrid CNN-LSTM architecture for continuous sign recognition

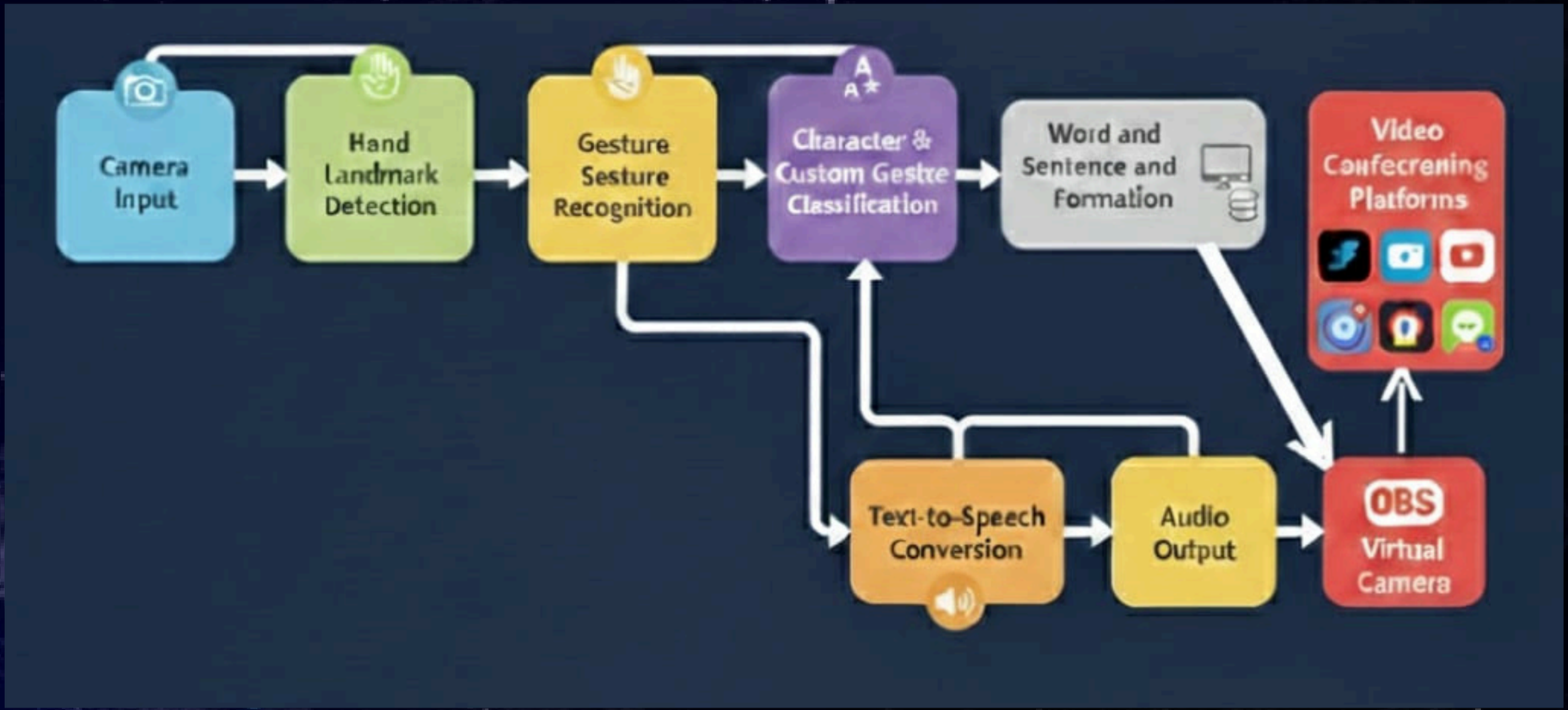
Output & Integration

- Text generation with sentence framing logic
- Text-to-Speech: pyttsx3 / gTTS
- OBS Virtual Camera for Zoom, Meet, and Teams

Workflow Diagram-1



Workflow Diagram-2



Uniqueness & Innovation

Gesture-Native Interaction Model

The system treats gestures as the primary interaction language, eliminating reliance on auxiliary inputs such as keyboards or manual corrections.

Continuity-Aware Recognition Pipeline

Gesture interpretation is stabilized across continuous motion sequences, reducing abrupt prediction changes common in frame-isolated systems.

Meeting-Embedded Translation Mechanism

Translation output is delivered as a native video stream, allowing sign-to-voice conversion to function directly inside live meetings instead of external windows.

User-Owned Data Boundary Design

All visual data remains within the user’s local environment, granting complete ownership and control of video feeds without third-party access.

Weekly Project Timeline

Week	Milestone	Apoorv	Rishit	Meenakshi	Deliverables
1	Project Initiation & Requirement Analysis	Scope definition	Technical feasibility	UI research	Approved problem statement, requirement specification
2	System Architecture & Design	EXE architecture design	Model architecture design	UI wireframes	System architecture diagram, UI wireframes
3	Dataset Preparation & Preprocessing	Data handling pipeline	Dataset collection & preprocessing	UI refinement	Preprocessed dataset, data pipeline
4	Model Development (Phase-I)	Integration planning	Initial model training	UI mockups	Baseline gesture recognition model
5	Model Optimization & Validation	EXE logic setup	Model tuning & validation	UI implementation	Optimized and validated ML model
6	System Integration	EXE bundling & integration	Model integration	UI integration	Functional sign-to-text system
7	Voice & OBS Integration	TTS and OBS pipeline	Performance optimization	UX enhancement	Sign-to-text-to-voice system
8	Testing & Quality Assurance	System testing	Accuracy evaluation	UI testing	Stable and tested application
9	Final Deployment & Documentation	Packaging & release	Final validation	Website & UI finalization	Final EXE, documentation, demo-ready system

Individual team member contributions

Member Name	Assigned Role	Primary Responsibilities	Technical Competencies	Time Commitment
Apoorv Mehrotra	EXE & System Integration Lead	System execution design, module integration, application packaging	Python, System Integration, EXE Packaging, OBS Studio	8 hours/week
Rishi Srivastava	Project Leader & Machine Learning Lead	Project coordination, model development, system evaluation	Machine Learning, Computer Vision, MediaPipe, Testing	10 hours/week
Meenakshi Gupta	UI/UX & Web Development Lead	Interface design, user experience optimization, web support	UI/UX Design, Frontend Development, Full-Stack	8 hours/week

Thank You !!